

Department Overview Department Name: Department of Mining Engineering Faculty: Faculty of Engineering & Architecture (established in 2004) Location: Takatu Campus (named after the Takatu Mountain), Quetta, Balochistan. History: The Department of Mining Engineering was established in 2008. Since its inception, a total of 8 batches have graduated and are serving across various public and private sector mining organizations. Accreditation: Fully accredited by the Pakistan Engineering Council (PEC). Core Focus: Cultivating technical competence and applied research to resolve real-time issues of the mining industry. The curriculum focuses heavily on environmentally friendly mineral extraction, engineering professional ethics, workplace safety, and sustainable development. Future Ambitions: Enhancing corporate and academic outreach by establishing sustainable linkages with local and international mining organizations.

Mission The department of Mining Engineering is striving for a high-quality undergraduate program supported by a compatible contemporary curriculum, applied research, and provision of technical competence to the graduates possessing skills to contribute to the Mining industry of Pakistan in general and Balochistan in particular with an environment-friendly approach.

Program Educational Objectives (PEOs) Civil and mining engineering graduates are expected to demonstrate the following attributes 3 to 5 years post-graduation:

1. Recognize, investigate, and address engineering problems related to the exploitation of mineral deposits of Pakistan/Balochistan by employing modern techniques.
2. Exhibit a high level of technical competence, applying research and problem-solving skills to produce solutions in the field of Mining Engineering.
3. Exercise professional excellence utilizing interpersonal skills focused on workplace safety, environmental sustainability, and professional engineering ethics.
4. Pursue lifelong learning by accepting practical challenges in the mining engineering field.
5. Bachelor of Science in Mining Engineering (BS Mining)

Admission Requirements Academic Background: F.Sc (Pre-Engineering) / ICS with Mathematics & Physics from a recognized board (or equivalent qualification) with at least 60% marks, OR a Diploma of Associate Engineer (DAE) in a relevant field with a minimum of 60% marks.

Degree Requirements • Total Credit Hours: 136 • Total Courses: 45 - 49 • Minimum Academic Standing: CGPA $\geq$ 2.0 • PLO Requirement: Mastery of the 12 expected standard engineering graduate attributes (Engineering Knowledge, Problem Analysis, Modern Tool Usage, etc.).

Curriculum Plan (Program Schema by Semester) Note: Credit hour structures denote (Theory + Lab).

1st Semester • MATHP120 | Analytic Geometry & Calculus | Theory CH: 3 | Lab CH: 0 • CHE-102 & 102L | Applied Chemistry | Theory CH: 3 | Lab CH: 1 • PHY-205 & 205L | Applied Physics | Theory CH: 3 | Lab CH: 1 • ENGG-201(MINE) & L | Engineering Drawing & Graphics | Theory CH: 1 | Lab CH: 2 • MINE-210 | Mining Engineering Fundamentals | Theory CH: 3 | Lab CH: 0 • Semester Total Credit Hours: 17

2nd Semester • MATHP-111 | Linear Algebra | Theory CH: 3 | Lab CH: 0 • ENGG-215 & 215L | Applied Thermodynamics | Theory CH: 3 | Lab CH: 1 • GEOE-201 & 201L | Applied Geology | Theory CH: 3 | Lab CH: 1 • HUM-101 | Islamic Studies / Ethics | Theory CH: 2 | Lab CH: 0 • HUM-364 / HUM-206 | English Composition & Comp. / Professional Ethics | Theory CH: 3/2 | Lab

CH: 0/0 • Semester Total Credit Hours: 18

3rd Semester • MINE-207 | Economic Minerals of Pakistan | Theory CH: 3 | Lab CH: 0 • ENGG-205 | Engineering Mechanics | Theory CH: 3 | Lab CH: 0 • ENGG-216 & 216L | Basic Electrical Technology | Theory CH: 3 | Lab CH: 1 • ENGG-206 & 206L | Fluid Mechanics | Theory CH: 3 | Lab CH: 1 • MATHA-214 | Differential Equations | Theory CH: 3 | Lab CH: 0 • Semester Total Credit Hours: 17

4th Semester • HUM-268 / CS-112L | Communication Skills / Computer Programming & Lab | Theory CH: 3/2 | Lab CH: 0/1 • MINE-209 & 209L | Mine Surveying | Theory CH: 3 | Lab CH: 1 • HUM-102 | Pakistan Studies | Theory CH: 2 | Lab CH: 0 • GEOE-206 & 206L | Mechanics of Material | Theory CH: 3 | Lab CH: 1 • STAT-102 | Probability & Statistics | Theory CH: 2 | Lab CH: 0 • Semester Total Credit Hours: 18

5th Semester • HUM-277 / MATHP-272 | Report Writing Skills / Numerical Method in Computing | Theory CH: 2/2 | Lab CH: 0/0 • GEOE-207 & 207L | Structural Geology | Theory CH: 3 | Lab CH: 1 • MINE-302 / MINE-307 & L | Mineral Exploration / Explosives Engineering | Theory CH: 3/3 | Lab CH: 0/1 • MGMT-201 | Entrepreneurship | Theory CH: 3 | Lab CH: 0 • Semester Total Credit Hours: 18

6th Semester • MINE-315 | Surface Mine Design | Theory CH: 3 | Lab CH: 0 • MINE-401 & 401L | Rock Mechanics | Theory CH: 3 | Lab CH: 1 • MINE-303 | Mineral Processing-I | Theory CH: 3 | Lab CH: 1 • TBA | Mine Power, Drainage & Material Handling | Theory CH: 3 | Lab CH: 0 • MINE-XXX | Computer Applications in Mining Engineering | Theory CH: 2 | Lab CH: 1 • Semester Total Credit Hours: 17

7th Semester • MINE-313 / MINE-316 & L | Mining Law / Mine Ventilation | Theory CH: 2/3 | Lab CH: 0/1 • MINE-310 & 310L | Mineral Processing-II | Theory CH: 3 | Lab CH: 1 • MINE-454L | Senior Design Project - I (FYDP-I) | Theory CH: 0 | Lab CH: 3 • ECON-105 / MGMT-304 | Engineering Economics / Organizational Behavior | Theory CH: 2/3 | Lab CH: 0/0 • Semester Total Credit Hours: 18

8th Semester • MINE-455L | Senior Design Project - II (FYDP-II) | Theory CH: 0 | Lab CH: 3 • MINE-317 / MINE-318 & L | Underground Mine Design / Mine Hazards & Safety | Theory CH: 3/3 | Lab CH: 0/1 • MGMT-211 / MINE-408 & L | Engineering Management / Tunnel Engineering | Theory CH: 2/3 | Lab CH: 0/1 • TBA | First Aid Course | Theory CH: 0 | Lab CH: 0 • Semester Total Credit Hours: 16

2. Master of Science Program (MS)

Admission Requirements Academic Background: 16 years of education or an equivalent benchmark qualification (such as a 4-year BS or B.E. degree) in a relevant engineering discipline from an HEC-recognized university with at least 60% marks (under the annual grading setup) or a minimum CGPA of 2.5 out of 4.0. (Note: The raw web copy contains a localized placeholder text errantly listing Chemical/Petroleum fields). Testing: A valid GAT General score with a passing benchmark of at least 50% marks.

Degree Requirements Total Credit Hours: 30 Total Coursework Requirements: 8 advanced postgraduate courses. Minimum Academic Standing: CGPA $\geq$ 2.5